1 The 1:1 Rule

ENG

1:1 — **MAP THE JOB, NOT THE SUBSTANCE.**

THE PART	A symbol compresses one electrical function into one glyph.
IN LIFE	Map each terminal, signal, and relation to one named cognitive role.
REGISTERS	A one-way electrical guard maps to one directed admissibility relation.
GUARD	1:1 means functional roles — not identical physics, parameters, or substrate.

2 Domains and Legend

ENG

P≻C≻E≻O — **FOUR TIERS · THREE LABELS.**

THE PART	Every board wire carries one tier and one labeling class.
IN LIFE	Precedence descends P $\succ$ C $\succ$ E $\succ$ O; O returns to P as labelled feedback.
REGISTERS	P principled · C cognitive · E evaluative · O operational.
GUARD	One badge per card. ISO = exact role match, ENG = stipulated rule or quantity, PED = teaching analogy.

3 Read a CAS State

ISO

4 AXES — **WHAT · WHERE · WHY · HOW.**

THE PART	A wire carries one signal with a name, a level, and a direction.
IN LIFE	Noetic (what) · Domain (where) · Drive (why) · Register (how).
REGISTERS	`^8^C_{F2}|A` = Cause, Cognitive, Wisdom, Analytical.
GUARD	Keep the four axes apart when you label a wire. Canonical Noetic names never change.

4 Operators and Firing

ENG

$+ \times |$ — **SPECIFY BEFORE YOU COMBINE.**

THE PART	A gate needs a defined threshold before it reads as true or false.
IN LIFE	$+$ is co-presence. $\times$ is generative integration. $|\langle \cdot \cdot \cdot \rangle$ specifies Registers.
REGISTERS	${}^{3}|A + {}^{3}|f$ keeps two manifestations distinct.
GUARD	Continuous states live on the open simplex. Use $\times$ only with a named emergent result.

5 Four Drift Stops

ISO

STOP — **REPULSION · COMPOSITION · TRIGGER · ARRIVAL.**

THE PART	A part does one job; a symbol asked to do two jobs is ambiguous.
IN LIFE	^{3} repels · ^{5} is accumulated composition · ^{6} decomposes · ^{8} triggers · ^{9} is brought to Mind.
REGISTERS	Reasoned landing may use ^{6} decomposition with ^{9} arrival.
GUARD	^{8} is not reasoning. ^{5} is not a container. Numbers on these cards are design choices, not measurements.

6 R · Resistor

ENG



LIMIT FLOW.

THE PART	Resistance limits current for a given voltage, and dissipates it as heat.
IN LIFE	Your brother says the thing designed to wound you. You say: pass the salt.
REGISTERS	$|A$ count to ten · $|f$ don't let him see · $|E$ the slow breath · $|T$ restraint nobody would know
GUARD	CAS resistance is a design scale, not measured human ohms.

7 C · Capacitor

ENG



RETAIN STATE.

THE PART	Capacitance holds charge across time and resists a change in voltage.
IN LIFE	Twenty years on, you can still hear exactly how she said your name.
REGISTERS	$|A$ a fact recalled · $|f$ warmth in a room · $|E$ shoulders softening · $|T$ part of who you are
GUARD	The Mind **is composed of** accumulated Ideas, not a container holding them.

8 L · Inductor

ENG

RESIST CHANGE.

THE PART	Inductance stores energy and opposes any change in current.
IN LIFE	Forty years of cracking a knuckle. Catching your hand mid-gesture feels like interrupting a sentence.
REGISTERS	$|A$ the old conclusion returning · $|f$ loyalty outgrown · $|E$ the hand before you notice · $|T$ a faith not to be hurried
GUARD	Define the repeating sequence; not every persistence effect is Rhythm.

9 Z · LC Tank

ENG



SELECT RESONANCE.

THE PART	L and C exchange energy; one frequency resonates most strongly.
IN LIFE	Nineteen years, one subject, and the argument ends the same way every time.
REGISTERS	$\|A$ the same three points · $\|f$ the subject that hurts · $\|E$ the room and the hour · $\|T$ what the years meant
GUARD	Change f_0 by changing L or C. Resolving to argue better is not a move.

10 D · Diode

ISO



ALLOW ONE DIRECTION.

THE PART	A PN junction conducts predominantly in one direction.
IN LIFE	You forgave him, and meant it. You also never lend him money again.
REGISTERS	$\|A$ what cannot be un-known · $\|f$ forgiveness without trust · $\|E$ a body that stays braced · $\|T$ a blessing, one way only
GUARD	Past reverse breakdown the gate conducts — hard, and not gently.

11 ZD · Zener

ENG



TRIP, THEN CLAMP.

THE PART	Reverse breakdown conducts at a threshold, then clamps.
IN LIFE	The criticism lands, your voice flattens, and you become extremely polite.
REGISTERS	$\|A$ the smooth agreeable sentence · $\|f$ flat and unreachable · $\|E$ the face that stops moving · $\|T$ protecting what you will not argue about
GUARD	The threshold must be reachable below every upstream ceiling.

12 T · Transformer

ENG



CHANGE LEVEL.

THE PART	Coupled windings trade voltage against current by a turns ratio.
IN LIFE	The same idea, three times in one day: to an eight-year-old, a colleague, your mother.
REGISTERS	$\|A$ three levels of abstraction · $\|f$ hard news said gently · $\|E$ voice and posture adjusted · $\|T$ meaning carried across a gulf
GUARD	A transformer changes level, never content — and it isolates the two sides.

13 PD · Photodiode

ENG

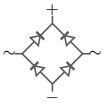


TURN STIMULUS INTO INTAKE.

THE PART	Incoming light produces current. A transducer, not a source.
IN LIFE	You walk into a cathedral for the architecture, and your eyes fill.
REGISTERS	$\|A$ a fact that changes what you knew · $\|f$ being moved · $\|E$ the hair lifting · $\|T$ awe at something you did not make
GUARD	Never `Photodiode<stimulus -> ^0>`. The output is a ^{1} intake.

14 BR · Bridge Rectifier

PED



NORMALIZE POLARITY.

THE PART	Four diodes make an alternating scalar polarity one-way.
IN LIFE	You cannot say whether the day was good or bad — only that it was loud.
REGISTERS	$\|A$ arousal without a label · $\|f$ keyed up, no direction · $\|E$ restless and prickling · $\|T$ no honest reading — hence PED
GUARD	Never convert ^{3} Negative repulsion into ^{2} Positive attraction.

15 Q · Cognitive Junction Transistor

ENG



BIAS SHAPES A TRANSITION.

THE PART	A control terminal governs a larger path: Cause / Bias / Effect.
IN LIFE	One sentence in the kitchen. One of you is irritated; the other is nine years old.
REGISTERS	$\|A$ two different conclusions · $\|f$ irritation vs a wound · $\|E$ the stomach dropping · $\|T$ a sentence that lands as calling
GUARD	Cause/Bias/Effect are cognitive roles, not universal BJT terminal identities.

16 U · Op-Amp

ENG



AMPLIFY A DIFFERENCE.

THE PART	High open-loop gain amplifies the difference of two inputs.
IN LIFE	“Nothing’s wrong” in a flat voice, and you build three weeks of theory.
REGISTERS	$\|A$ the theory · $\|f$ certainty you are unloved · $\|E$ hearing the quarter-tone · $\|T$ reading it as a verdict on your life
GUARD	Inputs must share one axis. Arithmetic minus is not Noetic involution.

17 Q/Q · Differential Pair

ENG



REMOVE WHAT BOTH SHARE.

THE PART	Matched devices amplify difference and reject common mode.
IN LIFE	Same redundancy, same meeting, same words. Eighteen months on, you differ.
REGISTERS	$\|A$ different conclusions · $\|f$ different depths of grief · $\|E$ the same cold, one shiver · $\|T$ one faith holds, one does not
GUARD	Folds Current Mirror. Inputs must be matched or you get noise.

18 Q · Active Load

ENG



TURN DIFFERENCE INTO OUTPUT.

THE PART	An active transistor load raises gain and forms a single-ended output.
IN LIFE	You have felt off all evening. One small remark, and you think: I'm leaving.
REGISTERS	$\|A$ suspicion becomes a decision · $\|f$ unease becomes resolve · $\|E$ reaching for the coat · $\|T$ wrongness becomes commitment
GUARD	A composite stage — not a new Noetic, and not a little person inside.

19 OSC · Oscillator

ENG



REPEAT ITSELF.

THE PART	Positive feedback sustains a periodic signal with no external drive.
IN LIFE	Two in the morning, the same four minutes, and not one new thought.
REGISTERS	$\|A$ rumination · $\|f$ a mood feeding itself · $\|E$ pacing · $\|T$ a question that returns nightly
GUARD	Name cycle, rate, gain, damping. Repetition alone is not validation.

20 Y · Crystal

ENG



HOLD THE BEAT.

THE PART	A high-Q resonator provides a precise, low-drift reference.
IN LIFE	Thirty years, the same walk, the same hour. When he missed it, people asked after him.
REGISTERS	$\|A$ a principle applied the same way · $\|f$ warmth unchanged on a bad day · $\|E$ a fixed daily practice · $\|T$ integrity others set clocks by
GUARD	A reference component, not a Noetic by identity. Steadiness can be exhausted.

21 PLL · Phase-Locked Loop

ISO



FOLLOW ANOTHER BEAT.

THE PART	A loop locks phase and frequency to a reference. Capture range, lock range.
IN LIFE	A month into a new job, you are taking breaks when your team takes breaks.
REGISTERS	$\|A$ absorbing a team's thinking · $\|f$ moods synchronising · $\|E$ falling into step · $\|T$ taking on what the group thinks matters
GUARD	State lock and capture range. Never claim instant entrainment.

22 U · ADC

ISO



SAMPLE AND CATEGORIZE.

THE PART	Sampling and quantization turn a continuous signal into discrete codes.
IN LIFE	Three weeks of "tired", and then you say it aloud: I think I'm lonely.
REGISTERS	$\|A$ finding the word · $\|f$ the feeling becoming bearable · $\|E$ noticing the clenched jaw · $\|T$ naming a longing as a question of meaning
GUARD	Quantizing is lossy. The word is not the feeling, and the code is not the signal.

23 U · DAC

ISO



RECONSTRUCT CONTINUOUS OUTPUT.

THE PART	Discrete codes become a smoothed continuous signal.
IN LIFE	"I am going to be honest with my father" — a sentence, then four years of behaviour.
REGISTERS	$\|A$ a decision becoming a plan · $\|f$ commitment as daily kindness · $\|E$ habit and routine · $\|T$ a conversion becoming a life's work
GUARD	Not the inverse of lossy ADC. Control in, behaviour out.

24 AND Gate

ENG



ALL REQUIRED.

THE PART	Output is true only when every input is true.
IN LIFE	You will lend him money if he has a plan AND has stopped drinking.
REGISTERS	$\|A$ two conditions · $\|f$ love and trust as separate flags · $\|E$ calm body and willing mind · $\|T$ conviction with practice
GUARD	Both-active is conflict, not confusion. Route it to deliberation.

25 OR Gate

ENG



ANY REQUIRED.

THE PART	Output is true when at least one input is true.
IN LIFE	Three signs mean leave the party. Any one of them is enough.
REGISTERS	$\|A$ any one of three reasons · $\|f$ any one feeling · $\|E$ any one warning sign · $\|T$ any one thing that breaks a commitment
GUARD	OR preserves alternatives; it does not integrate them.

26 NOT Gate

ENG



FLIP THE FLAG.

THE PART	Output is the Boolean complement of the input.
IN LIFE	"I'm not angry" — true, and both of you can hear something else.
REGISTERS	$\|A$ a threshold statement · $\|f$ equanimity read as coldness · $\|E$ stillness that is not calm · $\|T$ doubt that is not disbelief
GUARD	Involution only for $^{2} \leftrightarrow {}^{3}$, $^{5} \leftrightarrow {}^{6}$, $^{8} \leftrightarrow {}^{9}$. Self-duals stay themselves.

27 XOR Gate

ENG



EXACTLY ONE.

THE PART	Output is true when exactly one input is true.
IN LIFE	Two people told you something in confidence. You can honour one, not both.
REGISTERS	$\|A$ two conclusions, one action · $\|f$ two loyalties · $\|E$ two commitments at one hour · $\|T$ two callings
GUARD	XOR is **not** a both-active detector — XOR is false when both fire. Use AND. NAND/NOR invert; XNOR reports equality.

28 MUX · Multiplexer

ENG



CHOOSE ONE INPUT.

THE PART	A select line routes one of several inputs to one output.
IN LIFE	You have a version for your mother, a version for your team, a version for your oldest friend.
REGISTERS	$\|A$ which argument you reach for · $\|f$ which warmth appears · $\|E$ accent and posture · $\|T$ which self you bring to worship and to work
GUARD	Define every select value and a fail-closed default.

29 DMX · Demultiplexer

ENG



DISPATCH ONE INPUT.

THE PART	A select line sends one input to one of several outputs.
IN LIFE	One real insight. It has to reach your marriage, your work, or your father.
REGISTERS	$\|A$ one insight, one problem · $\|f$ one feeling, one relationship · $\|E$ one habit, one setting · $\|T$ one conviction, one domain
GUARD	Only the selected output receives it. Fan-out needs a different part.

30 K · Relay

ENG



ISOLATE CONTROL FROM CONSEQUENCE.

THE PART	A small control switches a much larger, electrically separate path.
IN LIFE	"I'm going to hand in my notice." One sentence; a mortgage changes state.
REGISTERS	$\|A$ one sentence, a career · $\|f$ a goodbye · $\|E$ a small movement, a large consequence · $\|T$ a vow that reorganises a life
GUARD	Cause controls Effect, never the reverse. Preserve isolation and fail state.

31 Schmitt Trigger

ENG



COMMIT, THEN HOLD.

THE PART	One threshold switches on; a lower threshold switches off.
IN LIFE	You decided on Tuesday, un-decided on Wednesday, and had re-decided eleven times by Friday.
REGISTERS	$\|A$ deciding once, not hourly · $\|f$ holding through a bad night · $\|E$ steady hands · $\|T$ a commitment that does not wobble with mood
GUARD	Name one scalar, both thresholds, and what would move the lower one.

32 REG · Voltage Regulator

ENG



RESTORE A BASELINE.

THE PART	Feedback holds output near a setpoint despite input and load changes.
IN LIFE	Four bad nights, and on the fourth you are short with everyone. Two good nights, and you are back.
REGISTERS	$\|A$ attention restored · $\|f$ mood returning to baseline · $\|E$ sleep, food, movement · $\|T$ a practice that keeps you oriented
GUARD	A regulator has a dropout. Past it there is nothing left to push with.

33 F · Fuse

ENG



CUT ONCE.

THE PART	Overcurrent melts an element and permanently opens the path.
IN LIFE	One rule, said aloud so that it cannot be renegotiated at three in the morning.
REGISTERS	$\|A$ a rule that ends it · $\|f$ the last time · $\|E$ leaving the building · $\|T$ a line that cannot be crossed without breaking faith
GUARD	Threshold must sit below the upstream clamp. Never cut the refusal path.

34 CB · Circuit Breaker

ENG



TRIP, THEN RESET SAFELY.

THE PART	Excess current opens a reusable protective switch.
IN LIFE	If the shouting starts, you leave the room and come back in twenty minutes.
REGISTERS	$\|A$ "tomorrow" · $\|f$ leaving the room · $\|E$ a walk mid-argument · $\|T$ a sabbath you can resume
GUARD	Define trip, latch, reset authority, and a fail-closed unreadable state.

35 RAIL · Drive Supply

ENG



SUPPLY MOTIVE POTENTIAL.

THE PART	A rail distributes supply potential to everything on the board.
IN LIFE	Eleven days without seeing anyone, and you cannot start anything. Nothing is wrong with your discipline.
REGISTERS	$\|A$ curiosity · $\|f$ connection · $\|E$ vitality · $\|T$ purpose
GUARD	A Drive is not voltage by identity. Do not score yourself out of seven.

36 REF · Reference Node

ENG

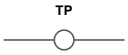


DEFINE ZERO.

THE PART	A reference node sets what the circuit calls zero.
IN LIFE	A quiet weekend at a friend's house, spent waiting for the thing that is about to happen.
REGISTERS	$\|A$ the assumption you reason from · $\|f$ what you call normal love · $\|E$ what your body calls normal tension · $\|T$ what you assume a life is for
GUARD	Not ^{0} Idea by identity. It is a convention, and it can be wrong.

37 TP · Probe

ISO



LOOK WITHOUT BECOMING THE SIGNAL.

THE PART	A test point exposes a conductor for measurement.
IN LIFE	You notice you are about to snap, you watch it happen, and you choose a different sentence.
REGISTERS	$\|A$ a thought seen as a thought · $\|f$ naming it live · $\|E$ the jaw, the breath · $\|T$ seeing what you live for, unedited
GUARD	A reading does not define, explain, or fix the state it reads.

38 SW · Switch / Enable

ENG



OPEN OR CLOSE A PATH.

THE PART	A control physically connects or disconnects one path.
IN LIFE	Not tonight. Sunday. On purpose — which is what keeps it from being avoidance.
REGISTERS	$\|A$ not thinking about it tonight · $\|f$ letting it go, for now · $\|E$ the phone in the other room · $\|T$ pausing a question for a season
GUARD	State open/closed meaning, default state, authority, and failure mode.

39 M · Meter / CAS-Ohm

ENG

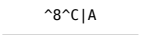


NAME WHAT YOU MEASURE.

THE PART	A meter reports a declared quantity by a declared method.
IN LIFE	"A seven out of ten." "What does seven mean?" You have no idea.
REGISTERS	$\|A$ ten minutes of concentration · $\|f$ a stated scale · $\|E$ hours slept · $\|T$ how often you acted on what you value
GUARD	Report the method with the number. A number without one is a feeling in a lab coat.

40 NET · Typed Label

ISO



NAME THE WHOLE SIGNAL.

THE PART	Identical net labels denote one electrical connection.
IN LIFE	You describe the week in pieces. She says: so you're frightened. You had not said that word.
REGISTERS	$\|A$ naming the belief · $\|f$ naming the feeling · $\|E$ naming the sensation · $\|T$ naming what you live for
GUARD	One label, one signal. A wrong label is worse than none.

41 Solid Arrow

ISO

MOVE FORWARD.

THE PART	A solid directed line marks the main forward path.
IN LIFE	“I did not sleep, and by Thursday I was snapping at people.”
REGISTERS	$\lvert A$ belief to conclusion · $\lvert f$ fear to withdrawal · $\lvert E$ tension to headache · $\lvert T$ commitment to a way of living
GUARD	Direction is flow, not attraction or preference.

42 Dashed Arrow

ISO

RETURN FEEDBACK.

THE PART	A dashed path marks a sampled return signal.
IN LIFE	The fifth time you have had this conversation. It is not about the topic.
REGISTERS	$\lvert A$ conclusion reshapes the premise · $\lvert f$ mood colours the memory · $\lvert E$ posture deepens the state · $\lvert T$ how you live changes what you believe
GUARD	Feedback is delayed and lossy. Name delay, gain, and sample.

43 C01 · CJT Atom

ENG

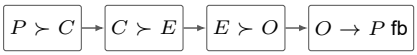


TRIGGER + COMPOSITION → ARRIVAL.

WHAT IT IS	A controlled transistor stage maps an input and a control to an output.
IN A LIFE	Someone says your name in a particular tone, and you are eight years old in a kitchen. The tone is the ^{8} trigger; the childhood is the ^{5} substrate.
THE CHAIN	A question triggers a learned procedure against training history.
GUARD	^{8} is associative triggering, not deductive reasoning.

44 C02 · Directed Domain Descent

ENG

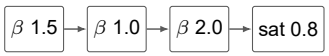


LICENSE P → C → E → O.

WHAT IT IS	Series direction guards with threshold protection control each crossing.
IN A LIFE	A conviction held in principle reaches your hands as a way of behaving — or it does not. Each crossing can happen or be refused.
THE CHAIN	A principle descends into reasoning, judgment, then action.
GUARD	Reverse RPM is lossy section–retraction, not ideal diode blocking.

45 C03 · Gain Cascade

ENG

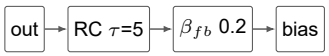


MULTIPLY DESIGN GAINS, RESPECT CLAMPS.

WHAT IT IS	Cascaded stages combine nominal gain but remain saturation-limited.
IN A LIFE	One small habit, repeated, becomes a disposition. The first honest sentence is effortful; the hundredth is character.
THE CHAIN	$1.5 \times 1.0 \times 2.0 = 3.0$ nominal; the final saturation is 0.8.
GUARD	Design arithmetic only. Measured Fisher–Rao and Jacobian gains are not interchangeable.

46 C04 · RC Feedback

ENG

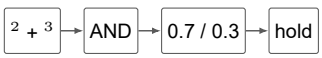


SMOOTH BEFORE RETURNING.

WHAT IT IS	An RC low-pass averages recent output before feedback.
IN A LIFE	You say something in a meeting, a face changes, and by your desk you have revised it into something safer.
THE CHAIN	$\tau = 5$ min and $\beta_{fb} = 0.2$ return a sampled trend.
GUARD	Feedback is delayed and bounded; it does not rewrite the output instantly.

47 C05 · Ambiguity Stabilizer

ENG

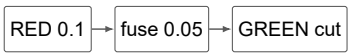


DETECT BOTH; HOLD THROUGH WOBBLE.

WHAT IT IS	An AND gate detects co-active flags; Schmitt hysteresis prevents chatter.
IN A LIFE	You want to go and you want to stay, both at once, and you are standing in the hallway. That is conflict, not confusion.
THE CHAIN	Conflict AND; on 0.7, off 0.3; middle values hold the prior state.
GUARD	XOR means exactly one, never both. Threshold continuous activation first.

48 C06 · CSB Red Path

ENG

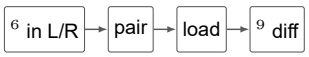


REFUSE, LATCH, CUT GREEN.

WHAT IT IS	A clamped RED path senses a reachable fuse that severs the alternate output.
IN A LIFE	One sentence — “I am going to say it tonight” — changes a marriage. The sentence took a second.
THE CHAIN	RED sat 0.1; fuse 0.05; the fuse cuts GREEN, not RED.
GUARD	$0.05 < 0.1$ is load-bearing. Saturation clamps; the fuse latches.

49 C07 · Differential Comparison

ENG

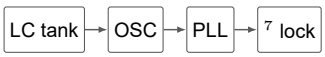


CANCEL COMMON; KEEP DIFFERENCE.

WHAT IT IS	Matched branches and a shared bias with an active load form a differential stage.
IN A LIFE	Two people hear the same criticism. One is fine and the other is not, and the difference is not the criticism.
THE CHAIN	Compare two argument activations on the same scale.
GUARD	One shared bias and matched gain are mandatory; comparison is not arbitrary subtraction.

50 C08 · Rhythm and Entrainment

ENG

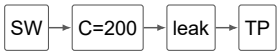


RESONATE, THEN LOCK.

WHAT IT IS	An LC tank selects resonance; a PLL locks it to a reference.
IN A LIFE	A practice rhythm and an external rhythm find each other: a baby's sleep and the household's week.
THE CHAIN	An internal practice rhythm gradually follows an external cadence.
GUARD	Crystal reference, resonance, and Rhythm are three distinct roles.

51 C09 · Memory Cell

ENG

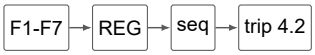


CHARGE, LEAK, REFRESH, SCAN.

WHAT IT IS	A switched capacitor retains charge, leaks, refreshes, and exposes a test point.
IN A LIFE	You remember, and you are changed by it, and over years the change softens. A low reading cannot tell you never-written from faded.
THE CHAIN	$C = 200$, leakage .001, refresh by a ^{7} cycle, scan below .05.
GUARD	Low charge alone cannot separate never-written from forgotten; keep provenance.

52 C10 · Drive Supply Board

ENG



POWER IN ORDER; FAIL TO SILENCE.

WHAT IT IS	Rails, regulators, a sequencer, brownout, and thermal cutoff manage supply.
IN A LIFE	Somebody is running on purpose and structure, has not slept in four months, and is still delivering — until the afternoon the lights go out.
THE CHAIN	Trip brownout 4.2, release 5.4; emergency cuts all CJTs.
GUARD	Operational output waits for P, C, and E power-good. Invalid supply gives silence.

53 How to Study the Deck

ENG

3 PASSES **SHAPE → JOB → STATE.**

THE PART	First name the conventional circuit function from the symbol.
IN LIFE	Then recite the cognitive function and the mapping class.
REGISTERS	Finally explain the example and the guardrail without looking.
GUARD	A remembered metaphor is not enough; preserve the role correspondence and the boundary.

54 Claim Boundary

ENG

BOUND **DESIGN REFERENCE, NOT MEASUREMENT.**

THE PART	The electrical definitions are standard functional summaries.
IN LIFE	The cognitive mappings are labelled ISO, ENG, or PED and need stated assumptions.
REGISTERS	Use these cards to design and to review candidate cognitive circuits.
GUARD	No card proves psychology, biology, AI internals, clinical effect, or production safety.

55 Print and Cut

ENG

CUT **ONE-SIDED STUDY CARDS.**

THE PART	Print at Actual Size on US Letter stock; the card border is the cut line.
IN LIFE	Keep the cards in number order, or sort them by circuit family.
REGISTERS	Seven sheets carry 56 cards: 5 guide, 29 part, 8 annotation, 10 circuit, 4 use.
GUARD	Do not use fit-to-page; scaling breaks text size and cut alignment. Varactor, Current mirror, NAND, NOR and XNOR are taught on their parent cards.

56 Build Your Own Circuit

ENG

WIRE **TYPE EVERY WIRE.**

THE PART	Name the source, terminal role, relation, control quantity, and destination.
IN LIFE	Label every state or scalar; then test the mapping class and the failure mode.
REGISTERS	Combine part cards into a circuit, then compare against cards 43–52.
GUARD	Check admissibility, thresholds, clamps, defaults, feedback, and reversibility first.